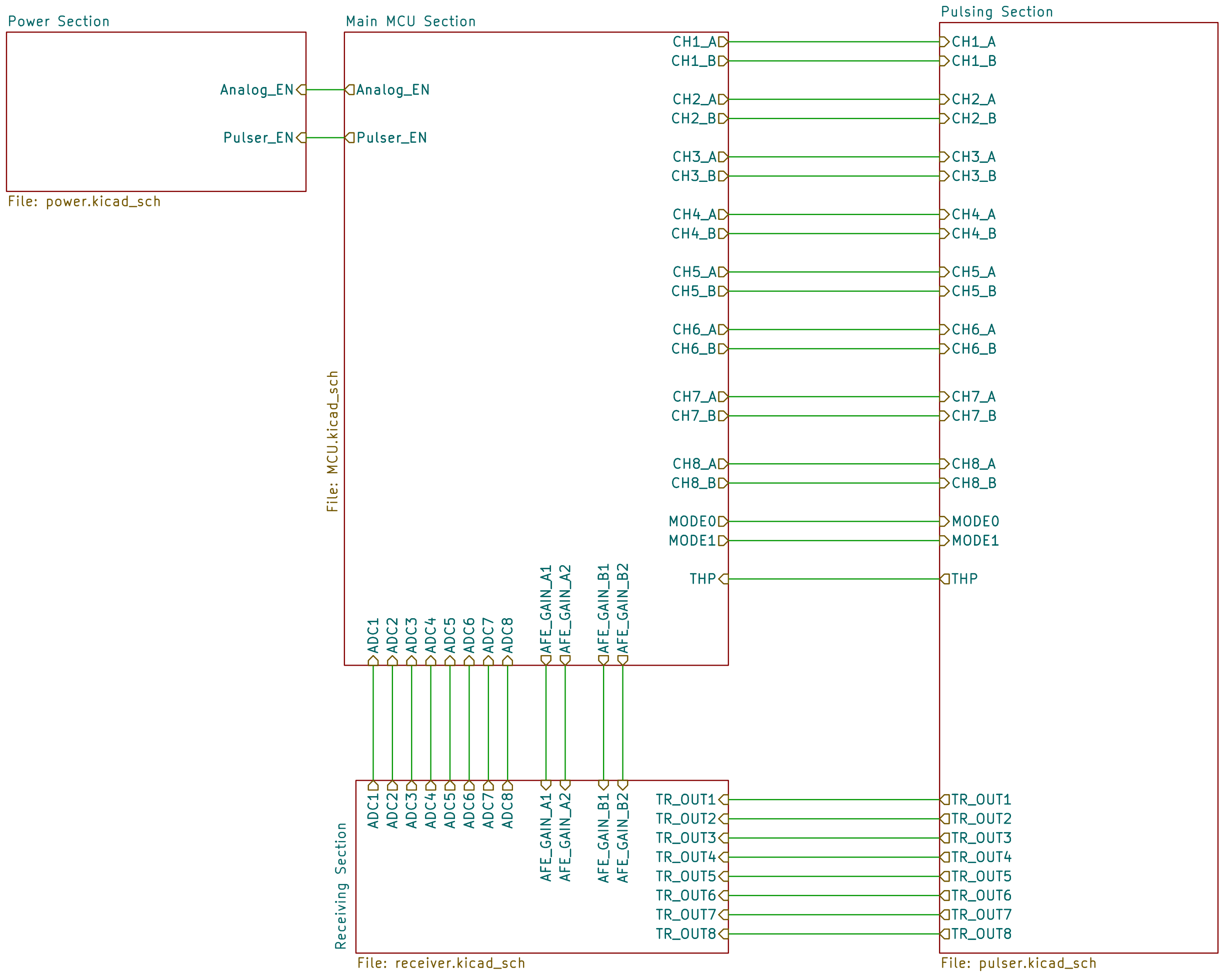


12 Volt (External) (**555mA**)	**7 Volt** (Buck Converter) (**350mA**)	**5 Volt** (LDO Regulator) (**200mA**)	LNA + VGA Supply (**160mA** – short circuit)
			Op-Amps (4 * 10mA = **40mA**)
		3.3 Volt (LDO Regulator) (**150mA**)	MCU Supply (**150mA**)
	5 Volt (Buck Converter) (**205mA**)	Positive Pulse Driver Supply (**100mA**)	
		3.3 Volt (LDO Regulator) (**5mA**)	Pulser Logic Supply (**5mA**)
		-5 Volt (Inverting Buck Converter) (**100mA**)	Negative Pulse Driver Supply (**100mA**)

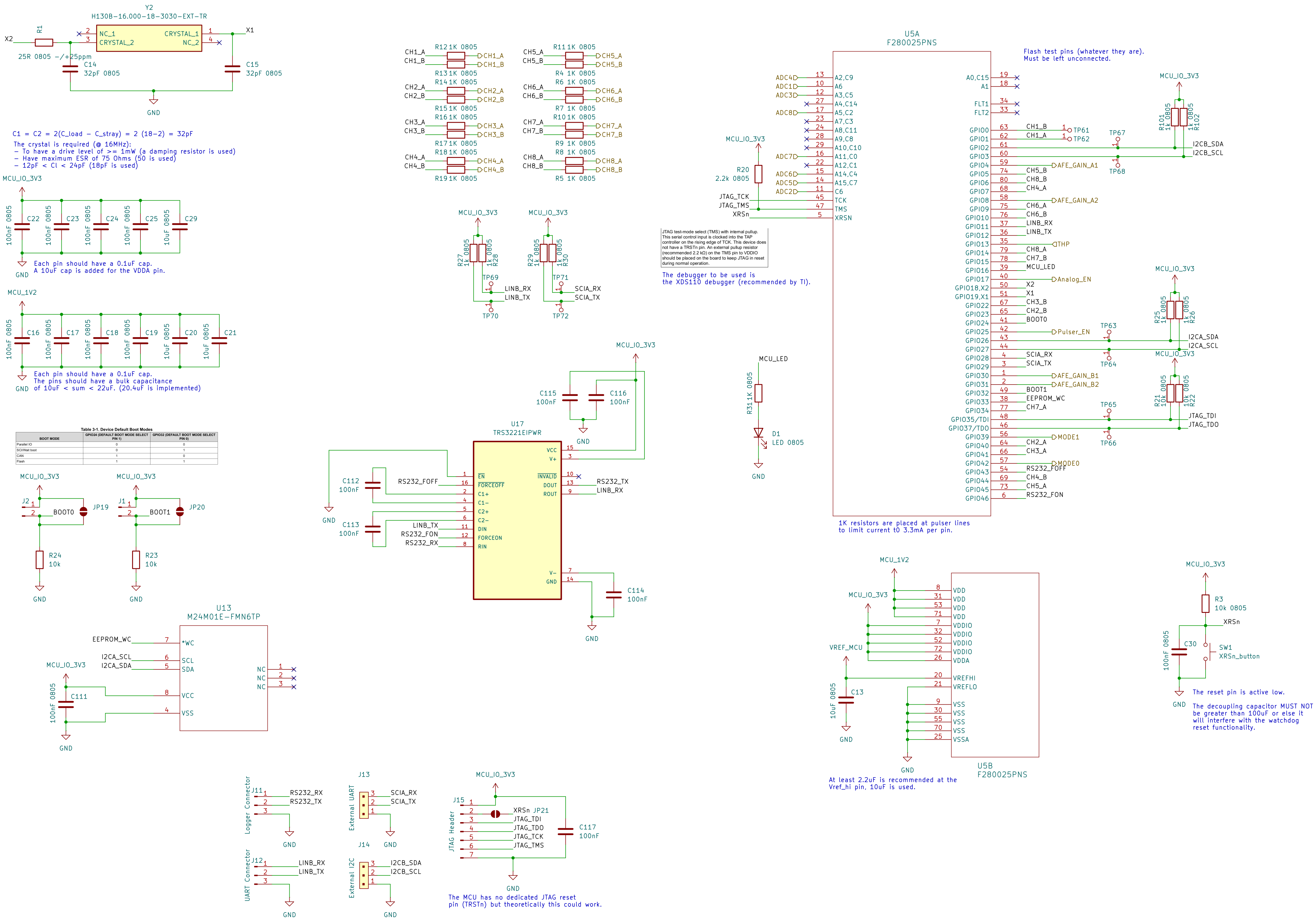
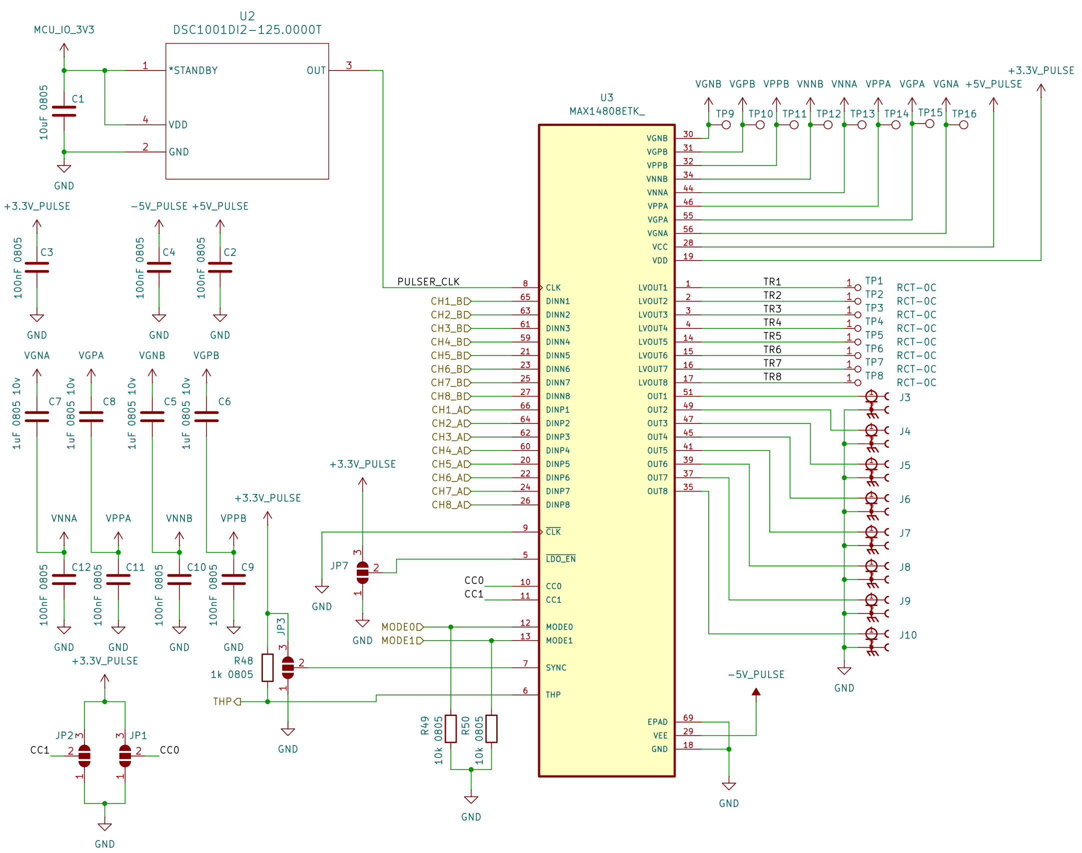


Table 3-1. Device Default Boot Modes

BOOT MODE	GPIO24 (DEFAULT BOOT MODE SELECT PIN 1)	GPIO32 (DEFAULT BOOT MODE SELECT PIN 0)
Parallel IO	0	0
SCI/Wait boot	0	1
CAN	1	0
Flash	1	1



VGXX: gate driver voltage supplies
(positive and negative for ports A & B)

VNNX/VPPX: positive and negative HV supplies
(for ports A & B)

Enabling/Disabling the internal floating power supplies (FPS):

requirements in most of the cases. Drive LDO_EN low or leave unconnected to enable the internal FPSs; drive LDO_EN high to disable the internal FPSs.

A triple solder jumper is placed
to either pull-up (select external FPS)
or pull-down (select internal FPS) the
LDOEN pin.

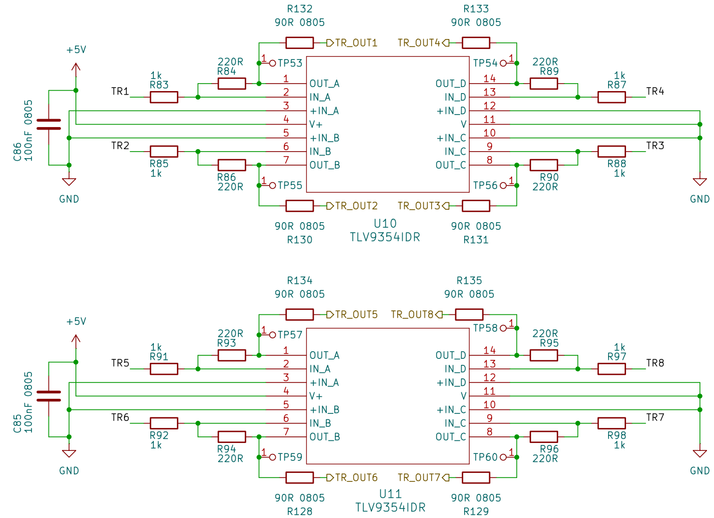
Table 5. Current Drive Selection

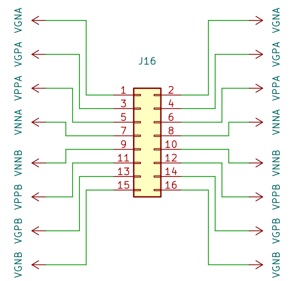
INPUTS		PULSER OUTPUT CURRENT (typ)
CC0	CC1	
0	0	2A
1	0	1.5A
0	1	1A
1	1	0.5A

Output swing from the pulser:
-1.2 -> 1.2 (2.4v swing)

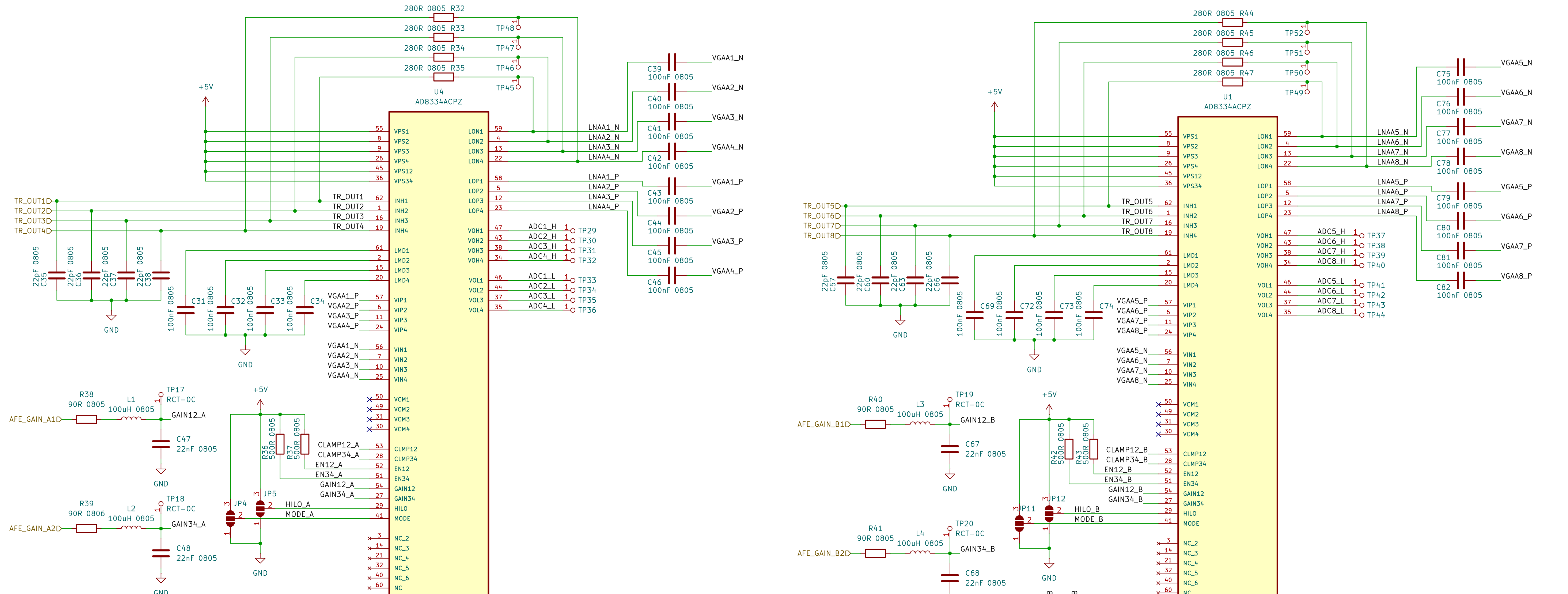
Input voltage range for the LNA+VGA:
-0.275 -> 0.275 (0.55v swing)

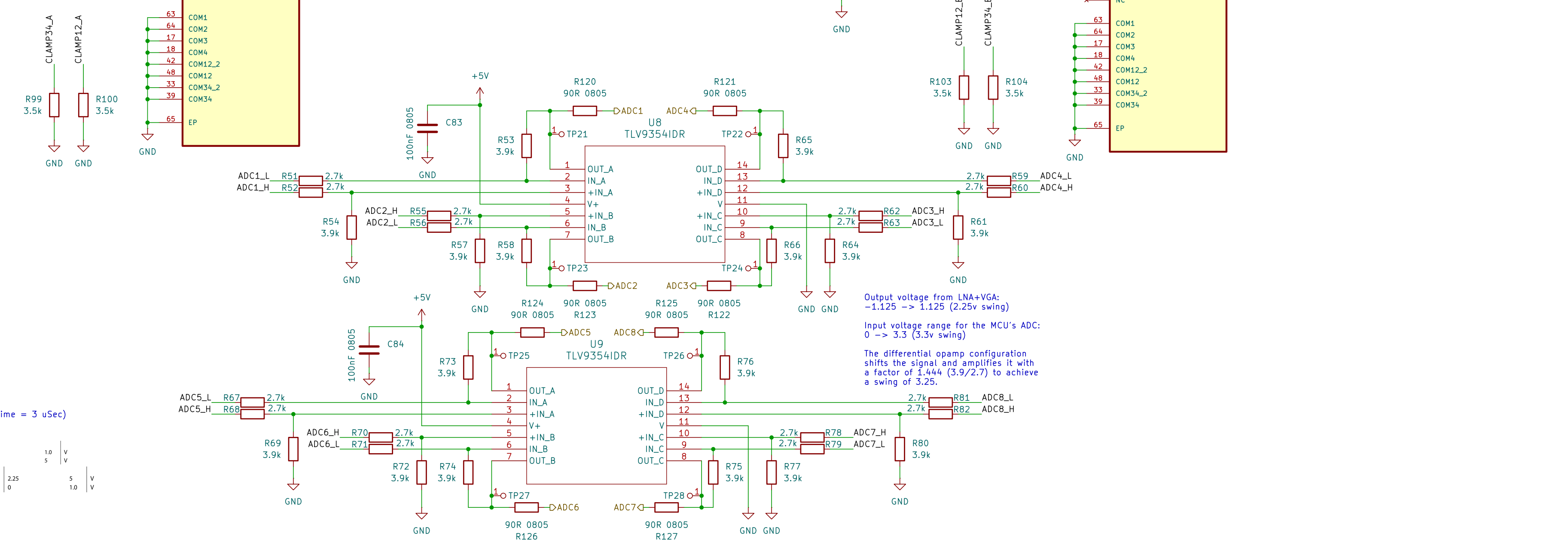
The differential opamp configuration
produces a gain of 0.22 for a swing
of 0.528v





External Power Connector





LNA Current					
AD8331 (ENBL)	Each channel	7.5	11	15	mA
AD8332, AD8334 (ENBL)	Each channel	7.5	12	15	mA
VGA Current					
AD8331 (ENBV)		7.5	14	20	mA
AD8332, AD8334 (ENBV)		7.5	17	20	mA

0.5k would draw 10mA, acceptable.

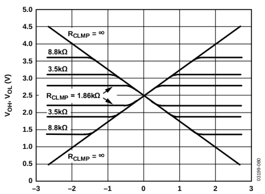


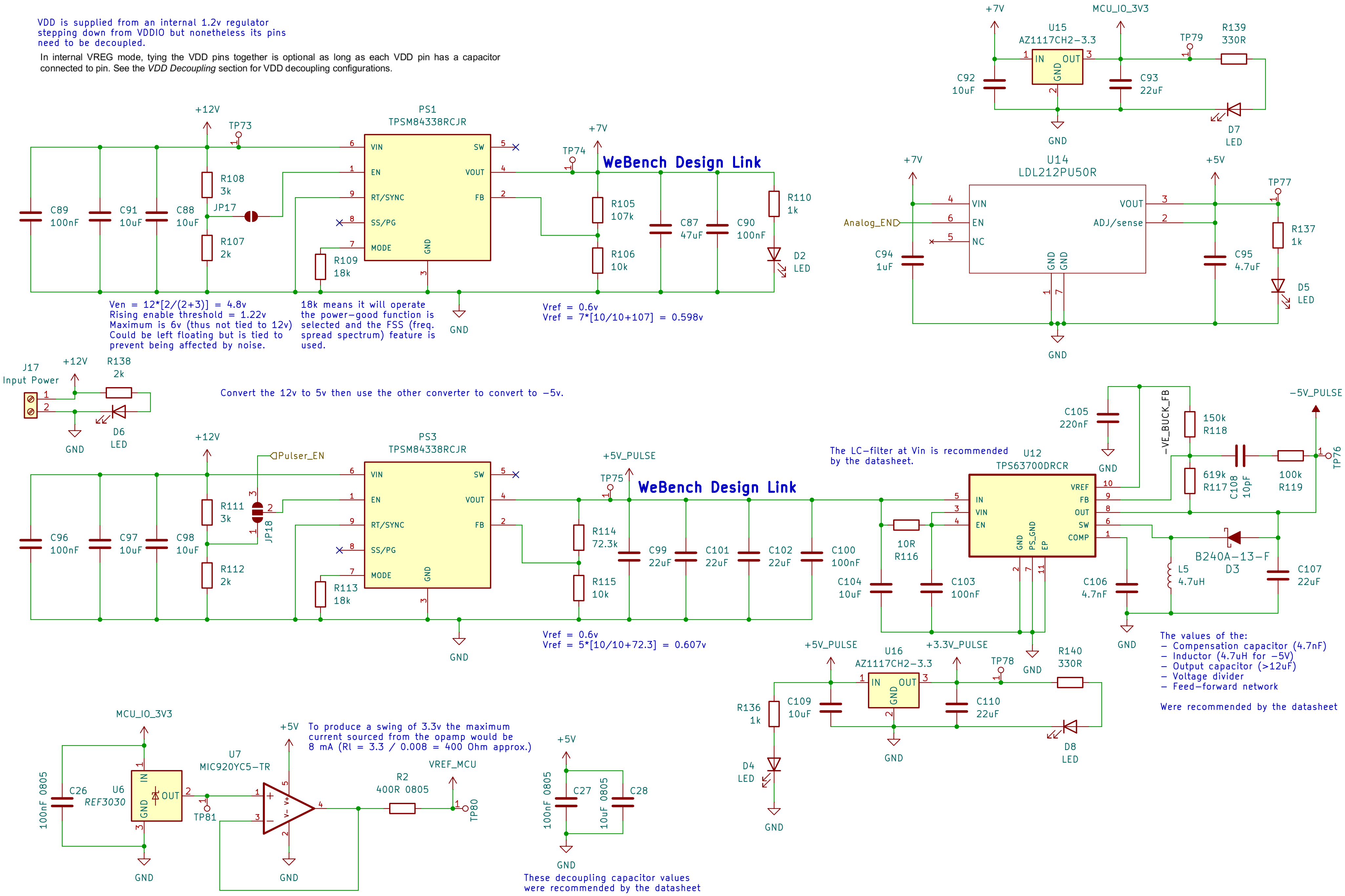
Figure 80. Output Clamping Characteristics

Table 7. LNA External Component Values for Common Source Impedances

R_{IN} (Ω)	R_{IZ} (Nearest STD 1% Value, Ω)	C_{SH} (pF)
50	280	22
75	412	12
100	562	8
200	1.13 k	1.2
500	3.01 k	None
6 k	∞	None

2nd order filter values were recommended
by a TI application note (spraa88a), specs:
- PWM frequency = 5 MHz
- Cut−off frequency = 107.3 KHz
- Damping ratio = 0.675 (over−damped but response time = 3 uSec)

MODE INTERFACE (PIN MODE)				
Logic Level for Positive Gain Slope		0	1.0	V
Logic Level for Negative Gain Slope		2.25	5	V
HILO GAIN RANGE INTERFACE (PIN HILO)				
Logic Level to Select HI Gain Range		2.25	5	V
Logic Level to Select LO Gain Range		0	1.0	V



VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.
In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.
+7V
MCU_IO_3V3
U15
AZ1117CH2-3.3
IN
OUT
GND
TP79
R139
330R
C92
10uF
C93
22uF
D7
LED
GND
+12V
TP73
PS1
TPSM84338RCJR
VIN
EN
RT/SYNC
SS/PG
MODE
SW
VOUT
FB
GND
TP74
WeBench Design Link
C89
100nF
C91
10uF
C88
10uF
R108
3k
JP17
R107
2k
R109
18k
R105
107k
R106
10k
C87
47uF
C90
100nF
R110
1k
D2
LED
GND
Ven = 12*[2/(2+3)] = 4.8v
Rising enable threshold = 1.22v
Maximum is 6v (thus not tied to 12v)
Could be left floating but is tied to prevent being affected by noise.
18k means it will operate the power-good function is selected and the FSS (freq. spread spectrum) feature is used.
Vref = 0.6v
Vref = 7*[10/10+107] = 0.598v
U14
LDL212PU50R
+5V
TP77
Analog_EN
VIN
EN
NC
VOUT
ADJ/sense
GND
C94
1uF
C95
4.7uF
R137
1k
D5
LED
GND
J17
Input Power
+12V
R138
2k
D6
LED
GND
Convert the 12v to 5v then use the other converter to convert to −5v.
PS3
TPSM84338RCJR
Pulser_EN
+5V_PULSE
TP75
WeBench Design Link
C96
100nF
C97
10uF
C98
10uF
R111
3k
JP18
R112
2k
R113
18k
R114
72.3k
R115
10k
C99
22uF
C101
22uF
C102
22uF
C100
100nF
GND
Vref = 0.6v
Vref = 5*[10/10+72.3] = 0.607v
The LC-filter at Vin is recommended by the datasheet.
U12
TPS63700DRCR
IN
VIN
EN
VREF
FB
OUT
SW
COMP
PS_GND
EP
R116
10R
C104
10uF
C103
100nF
C105
220nF
−VE_BUCK_FB
R118
150k
R117
619k
C108
10pF
R119
100k
−5V_PULSE
TP76
B240A-13-F
D3
L5
4.7uH
C106
4.7nF
C107
22uF
GND
The values of the:
− Compensation capacitor (4.7nF)
− Inductor (4.7uH for −5V)
− Output capacitor (>12uF)
− Voltage divider
− Feed−forward network
Were recommended by the datasheet
+5V_PULSE
U16
AZ1117CH2-3.3
+3.3V_PULSE
TP78
R140
330R
R136
1k
C109
10uF
C110
22uF
D4
LED
D8
LED
GND
MCU_IO_3V3
C26
100nF 0805
U6
REF3030
IN
OUT
GND
TP81
U7
MIC920YC5-TR
+5V
To produce a swing of 3.3v the maximum current sourced from the opamp would be 8 mA (Rl = 3.3 / 0.008 = 400 Ohm approx.)
R2
400R 0805
VREF_MCU
TP80
GND
+5V
C27
100nF 0805
C28
10uF 0805
GND
These decoupling capacitor values were recommended by the datasheet